

Jackie Lok

CURRICULUM VITAE

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Address: Sherrerd Hall, Charlton Street
ORFE Department
Princeton, NJ 08544, USA

EDUCATION

Princeton University, Princeton, NJ, USA
Ph.D. in Operations Research and Financial Engineering 2021–
Adviser: [Elizaveta Rebrova](#)

M.A. in Operations Research and Financial Engineering 2023

UNSW Sydney, Sydney, Australia
Bachelor of Science (Honours) in Pure Mathematics 2020
with First Class Honours and the University Medal
Supervisor: [Catherine Greenhill](#)
WAM: 97.16/100

Bachelor of Actuarial Studies (Co-op) in Mathematics with Distinction 2016–2019
WAM: 95.65/100

Wharton School, University of Pennsylvania, Philadelphia, PA, USA
International Exchange Semester, GPA: 4.00/4.00 2017

RESEARCH INTERESTS

My research is broadly in the mathematics of data science. I study the properties and behaviour of random mathematical structures and algorithms, and aim to use these insights to develop tools and models that allow us to better work with and understand large-scale, complex data.

My research interests mainly lie at the intersection of probability, statistics, and optimisation. More specifically, my interests include randomised algorithms, high-dimensional probability, random matrices, numerical linear algebra, and machine learning theory.

AWARDS & HONOURS

– Quad Fellowship, Schmidt Futures 2023

– Richard Stillwell '21 *24 and Agnes Newhall Stillwell Fellowship, Princeton University 2021

– University Medal in Pure Mathematics, UNSW Sydney 2020

– H.C. & M.E. Porter Memorial Scholarship, UNSW Sydney 2020

– The Faculty of Science Prize for Honours Year Science, UNSW Sydney 2020

– The George Szekeres Prize, UNSW Sydney 2019

– The Head of School's Prize, UNSW Sydney 2019

– UNSW Co-op Scholarship in Actuarial Studies 2016–2019

– UNSW Scientia Scholarship 2016–2019

– Harry Manson International Exchange Scholarship, UNSW Sydney 2017

Travel grants/awards

SIAM Student Travel Award (2024). ANU Annual Graduate School in Mathematical Aspects of Data Science Participant Support (2024). SEAS Travel Grant, Princeton University (2024, 2025).

PUBLICATIONS

Journal papers

- Roxanne He and Jackie Lok. “On Approximating the Potts Model with Contracting Glauber Dynamics”. *Advances in Applied Probability* (2025+). To appear. arXiv: [2404.18778](#).
- Jackie Lok and Elizaveta Rebrova. “A subspace constrained randomized Kaczmarz method for structure or external knowledge exploitation”. *Linear Algebra and its Applications* 698 (2024), pp. 220–260. DOI: [10.1016/j.laa.2024.06.010](#). arXiv: [2309.04889](#).

Conference papers

- Jackie Lok, Rishi Sonthalia, and Elizaveta Rebrova. “Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression”. *Proceedings of the 36th International Conference on Algorithmic Learning Theory (ALT 2025)*. Vol. 272. PMLR, 2025. arXiv: [2406.03696](#).

Preprints

- Jackie Lok and Elizaveta Rebrova. “Subspace-constrained randomized coordinate descent for linear systems with good low-rank matrix approximations” (2025). arXiv: [2506.09394](#).
- Rishi Sonthalia, Jackie Lok, and Elizaveta Rebrova. “On Regularization via Early Stopping for Least Squares Regression” (2024). arXiv: [2406.04425](#).

Miscellaneous

- Jackie Lok. *Markov chains, mixing times, and cutoff*. Honours thesis. 2020.

TALKS AND PRESENTATIONS

- The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, Canada: “Subspace-constrained Sketch-and-project Solvers for Linear Systems with Low-rank Structure” Jul 2025
- NLA group meeting, University of Oxford, UK: “Subspace-constrained sketch-and-project solvers for linear systems with low-rank structure” Jul 2025
- The 36th International Conference on Algorithmic Learning Theory (ALT 2025), Milan, Italy: “Error dynamics of mini-batch gradient descent with random reshuffling for least squares regression” Feb 2025
- Conference on the Mathematical Theory of Deep Neural Networks (DeepMath 2024), Philadelphia: “Error dynamics of mini-batch gradient descent with random reshuffling for least squares” (Poster) Nov 2024
- CUNY Graduate Center Harmonic Analysis & PDE Seminar: “A subspace constrained randomized Kaczmarz method” Nov 2024
- SIAM Conference on Mathematics of Data Science (MDS24), Atlanta: “A Subspace Constrained Randomized Kaczmarz Method for Structure or External Knowledge Exploitation” (Poster) Oct 2024
- NSF CompMath PI Meeting, University of Washington, Seattle: “A Subspace Constrained Randomized Kaczmarz Method for Structure or External Knowledge Exploitation” (Poster) Jul 2024
- Graduate student probability reading group, Princeton University: “Concentration for Random Matrix Products” Oct 2023
“Matrix Concentration Inequalities via the Method of Exchangeable Pairs” Nov 2022
- Honours presentation, UNSW Sydney: “Mixing times of Markov chains and the cutoff phenomenon” Nov 2020

TEACHING

Princeton University, *Princeton, NJ, USA*

Teaching Assistant (TA), ORFE Department

Responsible for delivering weekly precepts, holding office hours, grading problem sets and exams, and general course admin.

- [ORF 405: Regression and Applied Time Series](#) Fall 2025
- [ORF 387: Networks](#) Spring 2025
- [ORF 526: Probability Theory](#) Fall 2024
- [ORF 387: Networks](#) Spring 2024
- [ORF 363: Computing and Optimization](#) Fall 2023
- [ORF 350: Analysis of Big Data](#) Spring 2023
- [ORF 387: Networks](#) Fall 2022

University of Melbourne, *Melbourne, Australia*

Academic Tutor, School of Mathematics and Statistics

Responsible for delivering weekly tutorials and marking assessments.

- [MAST20004: Probability](#) Semester 1 2021

UNSW Sydney, *Sydney, Australia*

Academic Tutor, School of Risk and Actuarial Studies

Responsible for delivering weekly tutorials, marking exams and assessments, developing course materials, holding student consultations, and providing general course support.

- [ACTL3162: General Insurance Techniques](#) Term 3 2020
- [ACTL2102: Foundations of Actuarial Models](#) Term 2 2020
- [ACTL2111: Financial Mathematics for Actuaries](#) Term 1 2020
- [ACTL1101: Introduction to Actuarial Studies](#) Term 3 2019
- [ACTL2102: Foundations of Actuarial Models](#) Term 2 2019
- [ACTL3141: Actuarial Models and Statistics](#) Term 1 2019

OTHER ACTIVITIES

Mentoring

- [McGraw Graduate Teaching Fellow](#), *Princeton University* 2024–
- [ReMatch mentor](#), *Princeton University* 2023
- [ORFE Senior Thesis Writer's Group](#) co-leader, *Princeton University* 2022–2023
- [Drop-in Centre tutor](#), *School of Mathematics and Statistics, UNSW Sydney* 2020

Reviewing

IMA Journal of Numerical Analysis, Linear Algebra and its Applications, SIAM Journal on Matrix Analysis and Applications

WORK EXPERIENCE

icare, Actuarial Services Intern, *Sydney, Australia*

Aug 2018–Feb 2019

Supported the provision of actuarial advice and analysis for the NSW state insurer. Assisted with the reporting and valuation of outstanding claims liabilities, scenario analysis, preparation of financial budgets, claims experience monitoring, and the assessment of data quality and integrity.

Suncorp Group, Natural Perils Pricing Intern, *Sydney, Australia*

Feb 2018–Aug 2018

Collaborated in the research and development of a new natural peril pricing model in Python using analytical and machine learning techniques with insurance and geospatial datasets. Developed interactive tool using SAS and Python to identify and visualise exposure concentration risks as part of an automated monitoring pipeline.

MetLife Australia, Capital and Valuation Intern, *Sydney, Australia*

Nov 2016–Feb 2017

Assisted with financial reporting, reserving and scenario analysis for group life insurance.

LANGUAGES AND SKILLS

Languages

English (native), Cantonese (fluent), Mandarin (beginner), German (beginner).

Computing

- Proficient with Python. Experience with other programming languages including Julia, R, Java, MATLAB, SQL, and SAS.
- Competent with L^AT_EX.
- Experience with Microsoft Excel, Word, and PowerPoint.

Online courses

- [Probabilistic Graphical Models Specialization](#) (Coursera, Stanford University) 2021
- [Deep Learning Specialization](#) (Coursera, DeepLearning.AI) 2021
- [Machine Learning](#) (Coursera, Stanford University) 2018